

MODULE 3

Functional Dependency

Normal Forms

Multi-valued Dependency

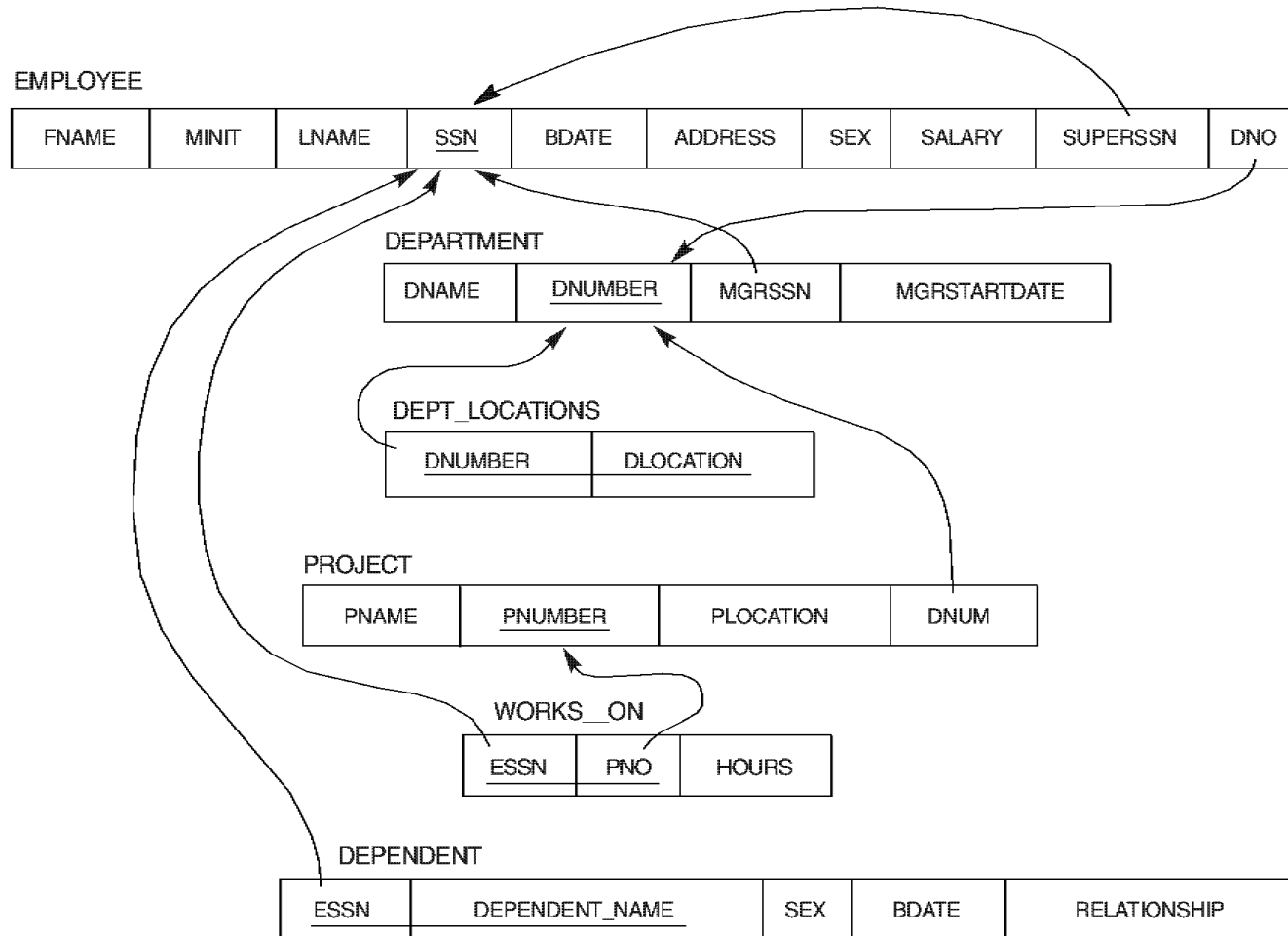
Join dependency

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Sources:

Pearson Education, Inc. 2011, Elmasri/Navathe, Fundamentals of Database Systems, Sixth Edition

Figure 5. Referential integrity constraints displayed on the COMPANY relational database schema diagram.



1.1 Semantics of the Relation Attributes

GUIDELINE 1: Informally, each tuple in a relation should represent one entity or relationship instance.

- ❑ Mixing attributes of multiple entities may cause problems
- ❑ Information is stored redundantly wasting storage
- ❑ Problems with update anomalies
 - ❑ Insertion anomalies
 - ❑ Deletion anomalies
 - ❑ Modification anomalies

Example states for EMP_DEPT and EMP_PROJ resulting from applying NATURAL JOIN

EMP_DEPT					redundancy	
ENAME	<u>SSN</u>	BDATE	ADDRESS	DNUMBER	DNAME	DMGRSSN
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

EXAMPLE OF AN UPDATE ANOMALY (1)

Consider the relation:

EMP_PROJ (Emp#, Proj#, Ename, Pname, No_hours)

- **Update Anomaly:** Changing the name of project number P1 from “Billing” to “Customer-Accounting” may cause this update to be made for all 100 employees working on project P1.

EXAMPLE OF AN UPDATE ANOMALY (2)

- **Insert Anomaly:** Cannot insert a project unless an employee is assigned to .
Inversely - Cannot insert an employee unless an he/she is assigned to a project.
- **Delete Anomaly:** When a project is deleted, it will result in deleting all the employees who work on that project. Alternately, if an employee is the sole employee on a project, deleting that employee would result in deleting the corresponding project.

Guideline to Redundant Information in Tuples and Update Anomalies

- ❑ **GUIDELINE 2:** Design a schema that does not suffer from the insertion, deletion and update anomalies. If there are any present, then note them so that applications can be made to take them into account

1.3 Null Values in Tuples

GUIDELINE 3: Relations should be designed such that their tuples will have as few NULL values as possible

- Reasons for nulls:
 - ▣ attribute not applicable or invalid
 - ▣ attribute value unknown (may exist)
 - ▣ value known to exist, but unavailable
- Attributes that are NULL frequently could be placed in separate relations (with the primary key)

1.4 Spurious Tuples

- ❑ Bad designs for a relational database may result in erroneous results for certain JOIN operations
- ❑ The "lossless join" property is used to guarantee meaningful results for join operations

GUIDELINE 4: The relations should be designed to satisfy the lossless join condition. No spurious tuples should be generated by doing a natural-join of any relations.

EMP_LOCS

ENAME	PLOCATION
Smith, John B.	Bellaire
Smith, John B.	Sugarland
Narayan, Ramesh K.	Houston
English, Joyce A.	Bellaire
English, Joyce A.	Sugarland
Wong, Franklin T.	Sugarland
Wong, Franklin T.	Houston
Wong, Franklin T.	Stafford

Zelaya, Alicia J.	Stafford
Jabbar, Ahmad V.	Stafford
Wallace, Jennifer S.	Stafford
Wallace, Jennifer S.	Houston
Borg,James E.	Houston

EMP_PROJ1

SSN	PNUMBER	HOURS	PNAME	PLOCATION
123456789	1	32.5	Product X	Bellaire
123456789	2	7.5	Product Y	Sugarland
666884444	3	40.0	Product Z	Houston
453453453	1	20.0	Product X	Bellaire
453453453	2	20.0	Product Y	Sugarland
333445555	2	10.0	Product Y	Sugarland
333445555	3	10.0	Product Z	Houston
333445555	10	10.0	Computerization	Stafford
333445555	20	10.0	Reorganization	Houston

999887777	30	30.0	Newbenefits	Stafford
999887777	10	10.0	Computerization	Stafford
987987987	10	35.0	Computerization	Stafford
987987987	30	5.0	Newbenefits	Stafford
987654321	30	20.0	Newbenefits	Stafford
987654321	20	15.0	Reorganization	Houston
888665555	20	null	Reorganization	Houston

Result of applying NATURAL JOIN to the tuples above the dotted lines in
EMP_PROJ1 and EMP_LOCS
(2, 4, 5, 7, 8, 10, 12, 13, 14, 16, 19)

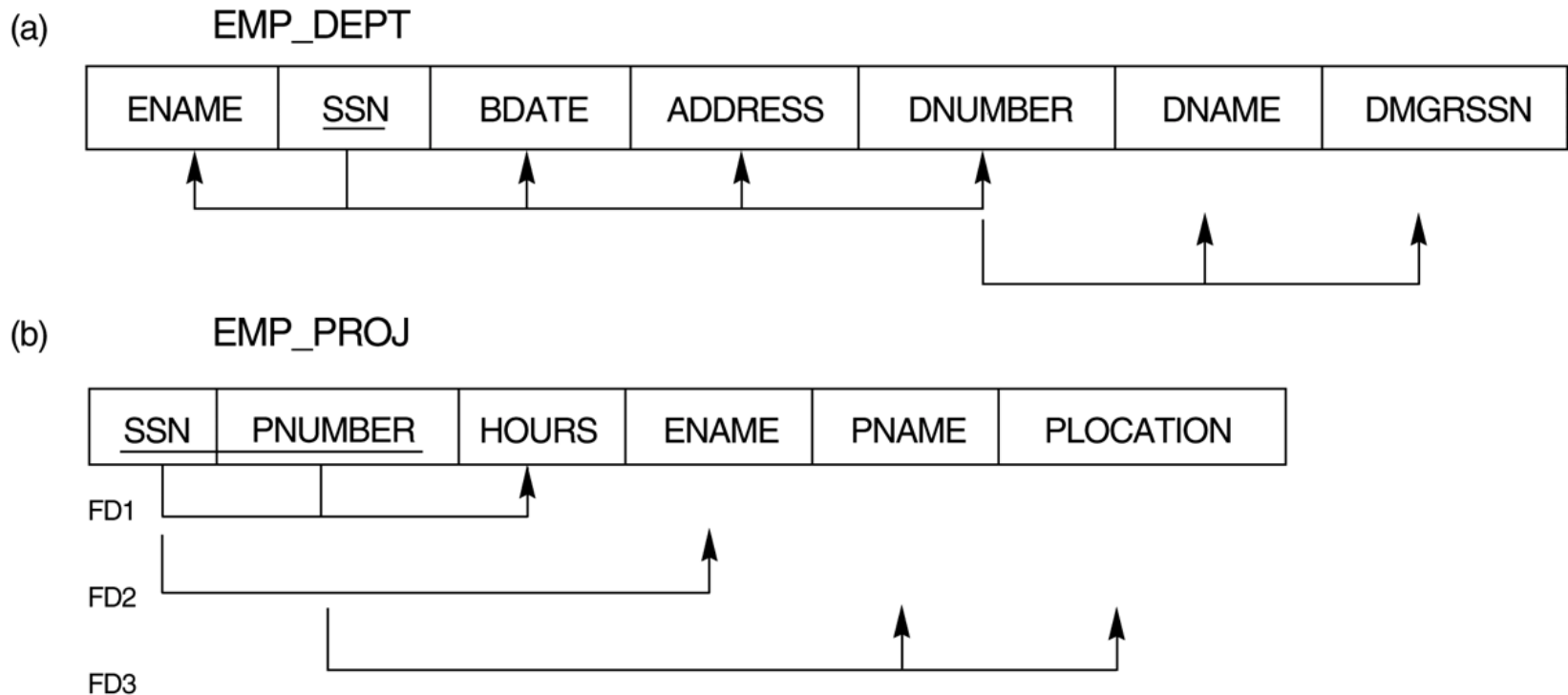
SSN	PNUMBER	HOURS	PNAME	PLOCATION	
123456789	1	32.5	ProductX	Bellaire	Smith,John B.
123456789	1	32.5	ProductX	Bellaire	English,Joyce A.
123456789	2	7.5	ProductY	Sugarland	Smith,John B.
123456789	2	7.5	ProductY	Sugarland	English,Joyce A.
123456789	2	7.5	ProductY	Sugarland	Wong,Franklin T.
666884444	3	40.0	ProductZ	Houston	Narayan,Ramesh K.
666884444	3	40.0	ProductZ	Houston	Wong,Franklin T.
453453453	1	20.0	ProductX	Bellaire	Smith,John B.
453453453	1	20.0	ProductX	Bellaire	English,Joyce A.
453453453	2	20.0	ProductY	Sugarland	Smith,John B.
453453453	2	20.0	ProductY	Sugarland	English,Joyce A.
453453453	2	20.0	ProductY	Sugarland	Wong,Franklin T.
333445555	2	10.0	ProductY	Sugarland	Smith,John B.
333445555	2	10.0	ProductY	Sugarland	English,Joyce A.
333445555	2	10.0	ProductY	Sugarland	Wong,Franklin T.
333445555	3	10.0	ProductZ	Houston	Narayan,Ramesh K.
333445555	3	10.0	ProductZ	Houston	Wong,Franklin T.
333445555	10	10.0	Computerization	Stafford	Wong,Franklin T.
333445555	20	10.0	Reorganization	Houston	Narayan,Ramesh K.
333445555	20	10.0	Reorganization	Houston	Wong,Franklin T.

⋮

2.1 Functional Dependencies (1)

- A set of attributes X *functionally determines* a set of attributes Y if the value of X determines a unique value for Y
- $X \rightarrow Y$ (X determines Y) holds if whenever two tuples have the same value for X , they *must have* the same value for Y
- For any two tuples t_1 and t_2 in any relation instance $r(R)$: *If* $t_1[X]=t_2[X]$, *then* $t_1[Y]=t_2[Y]$
- $X \rightarrow Y$ in R specifies a *constraint* on all relation instances $r(R)$

Two relation schemas suffering from update anomalies.



Examples of FD constraints

- social security number determines employee name

SSN $\rightarrow$ ENAME

- project number determines project name and location

PNUMBER $\rightarrow$ {PNAME, PLOCATION}

- employee ssn and project number determines the hours per week that the employee works on the project

{SSN, PNUMBER} $\rightarrow$ HOURS

Examples of FD constraints

- An FD is a property of the attributes in the schema R
- The constraint must hold on *every relation instance* $r(R)$
- If K is a key of R , then K functionally determines all attributes in R (since we never have two distinct tuples with $t_1[K]=t_2[K]$)

2.2 Inference Rules for FDs

- Given a set of FDs F , we can *infer* additional FDs that hold whenever the FDs in F hold

Armstrong's inference rules:

IR1. (**Reflexive**) If Y subset-of X , then $X \rightarrow Y$

IR2. (**Augmentation**) If $X \rightarrow Y$, then $XZ \rightarrow YZ$

(Notation: XZ stands for $X \cup Z$)

IR3. (**Transitive**) If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

- IR1, IR2, IR3 form a *sound* and *complete* set of inference rules

Inference Rules for FDs (2)

Some **additional inference rules** that are useful:

(Decomposition) If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$

(Union) If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

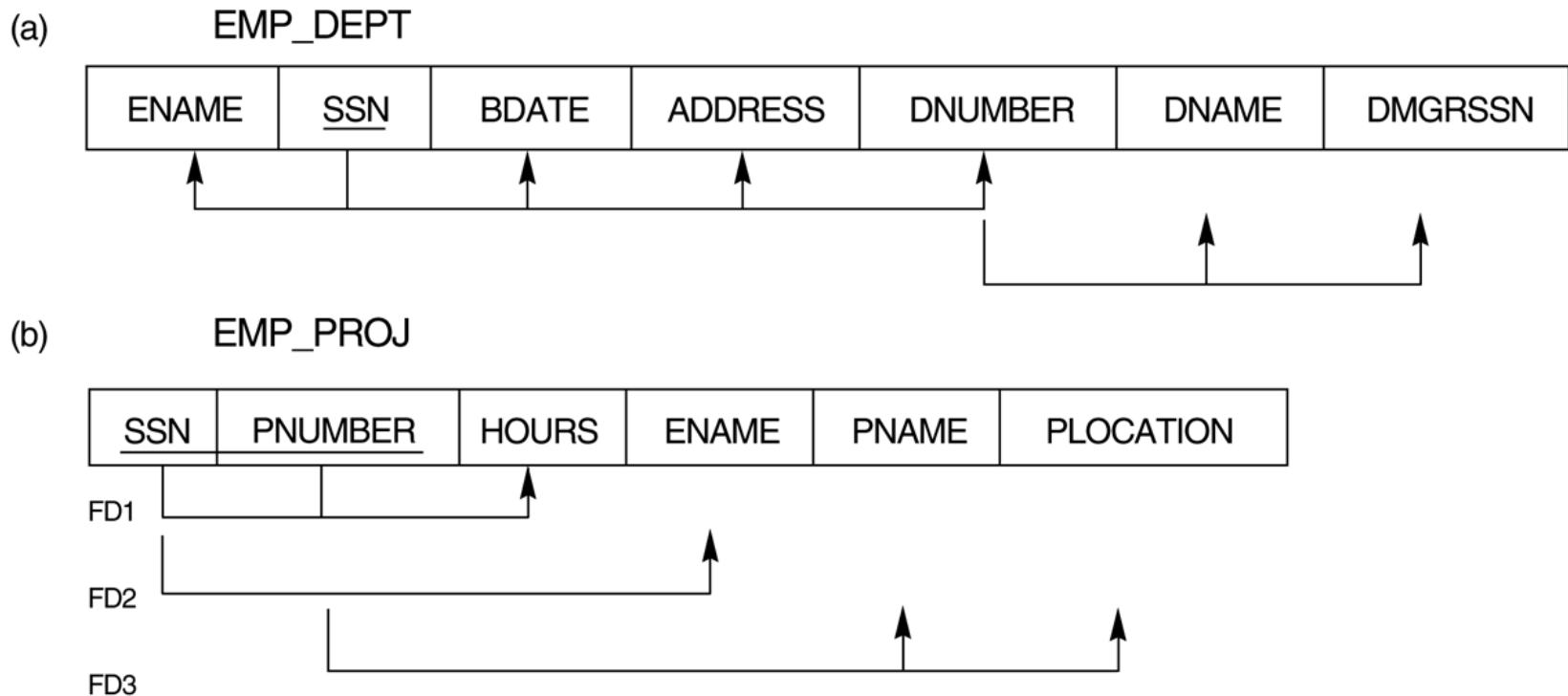
(Pseudotransitivity) If $X \rightarrow Y$ and $WY \rightarrow Z$, then $WX \rightarrow Z$

□ The last three inference rules, as well as any other inference rules, can be deduced from IR1, IR2, and IR3 (completeness property)

Inference Rules for FDs (3)

- **Closure** of a set F of FDs is the set F^+ of all FDs that can be inferred from F
- **Closure** of a set of attributes X with respect to F is the set X^+ of all attributes that are functionally determined by X
- X^+ can be calculated by repeatedly applying IR1, IR2, IR3 using the FDs in F

Two relation schemas suffering from update anomalies.



Inference Rules for FDs (4)

- $F = \{\text{SSN} \rightarrow \text{ENAME},$
 $\text{PNUMBER} \rightarrow \{\text{PNAME}, \text{PLOCATION}\}$
 $\{\text{SSN}, \text{PNUMBER}\} \rightarrow \text{HOURS}\}$
- $\{\text{SSN}\}^+ = \{\text{SSN}, \text{ENAME}\}$
 $\{\text{PNUMBER}\}^+ = \{\text{PNUMBER}, \text{PNAME},$
 $\text{PLOCATION}\}$
 $\{\text{SSN}, \text{PNUMBER}\}^+ = \{\text{SSN}, \text{ENAME}, \text{PNUMBER},$
 $\text{PNAME}, \text{PLOCATION}, \text{HOURS}\}$

2.3 Equivalence of Sets of FDs

□ Two sets of FDs F and G are **equivalent** if:

- every FD in F can be inferred from G , *and*
- every FD in G can be inferred from F

□ Hence, F and G are equivalent if $F^+ = G^+$

Definition: F **covers** G if every FD in G can be inferred from F (i.e., if G^+ is subset-of F^+)

□ F and G are equivalent if F covers G and G covers F .

2.4 Minimal Sets of FDs (1)

□ A set of FDs is **minimal** if it satisfies the following conditions:

- (1) Every dependency in F has a single attribute for its RHS.
- (2) We cannot remove any dependency from F and have a set of dependencies that is equivalent to F .
- (3) We cannot replace any dependency $X \rightarrow A$ in F with a dependency $Y \rightarrow A$, where Y proper-subset-of X (Y subset-of X) and still have a set of dependencies that is equivalent to F .

Minimal Sets of FDs (2)

- Every set of FDs has an equivalent minimal set
- There can be several equivalent minimal sets
- There is no simple algorithm for computing a minimal set of FDs that is equivalent to a set F of FDs
- To synthesize a set of relations, we assume that we start with a set of dependencies that is a minimal set

Normalization of Relations (1)

- **Normalization:** The process of decomposing unsatisfactory "bad" relations by breaking up their attributes into smaller relations
- **Normal form:** Condition using keys and FDs of a relation to certify whether a relation schema is in a particular normal form

3.2 Practical Use of Normal Forms

- **Normalization** is carried out in practice so that the resulting designs are of high quality and meet the desirable properties
- The practical utility of these normal forms becomes questionable when the constraints on which they are based are **hard to understand** or to **detect**
- The database designers ***need not*** normalize to the highest possible normal form. (usually up to 3NF, BCNF or 4NF)
- **Denormalization:** the process of storing the join of higher normal form relations as a base relation—which is in a lower normal form

Definitions of Keys and Attributes Participating in Keys

- A **superkey** of a relation schema $R = \{A_1, A_2, \dots, A_n\}$ is a set of attributes S subset-of R with the property that no two tuples t_1 and t_2 in any legal relation state r of R will have $t_1[S] = t_2[S]$
- A **key** K is a superkey with the *additional property* that removal of any attribute from K will cause K not to be a superkey any more.

Definitions of Keys and Attributes

Participating in Keys (2)

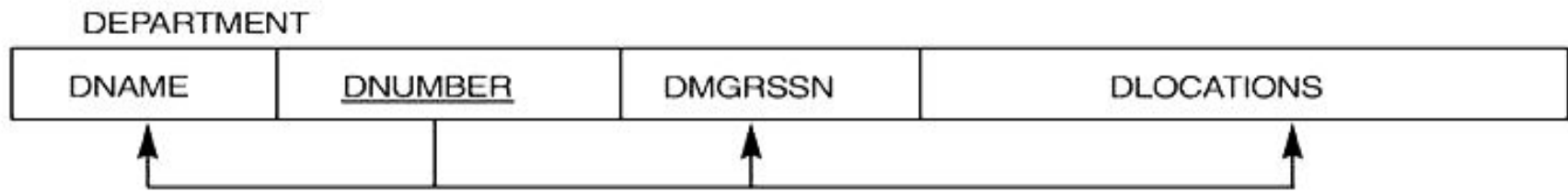
- If a relation schema has more than one key, each is called a **candidate key**. One of the candidate keys is *arbitrarily* designated to be the **primary key**, and the others are called *secondary keys*.
- A **Prime attribute** must be a member of *some candidate key*
- A **Nonprime attribute** is not a prime attribute—that is, it is not a member of any candidate key.

3.2 First Normal Form

- ❑ Disallows composite attributes, multivalued attributes, and **nested relations**; attributes whose values *for an individual tuple* are non-atomic
- ❑ Considered to be part of the definition of relation

Normalization into 1NF

(a)



(b)

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

(c)

DEPARTMENT

DNAME	<u>DNUMBER</u>	DMGRSSN	<u>DLOCATION</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

Second Normal Form

- Uses the concepts of **FDs**, **primary key**

Definitions:

- **Prime attribute** - attribute that is member of the primary key K
- **Full functional dependency** - a FD $Y \rightarrow Z$ where removal of any attribute from Y means the FD does not hold any more

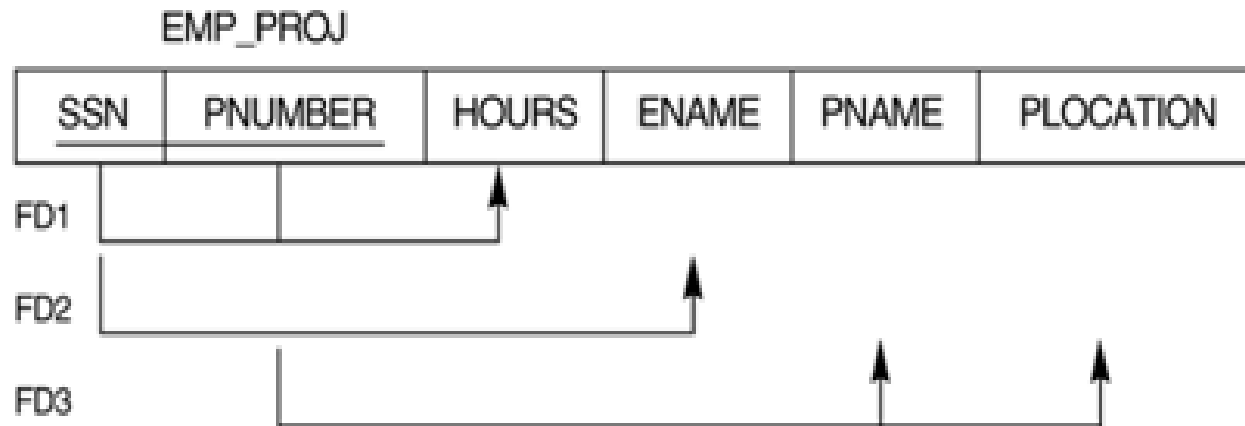
Examples:

- $\{SSN, PNUMBER\} \rightarrow HOURS$ is a full FD since neither $SSN \rightarrow HOURS$ nor $PNUMBER \rightarrow HOURS$ hold
- $\{SSN, PNUMBER\} \rightarrow ENAME$ is *not* a full FD (it is called a *partial dependency*) since $SSN \rightarrow ENAME$ also holds

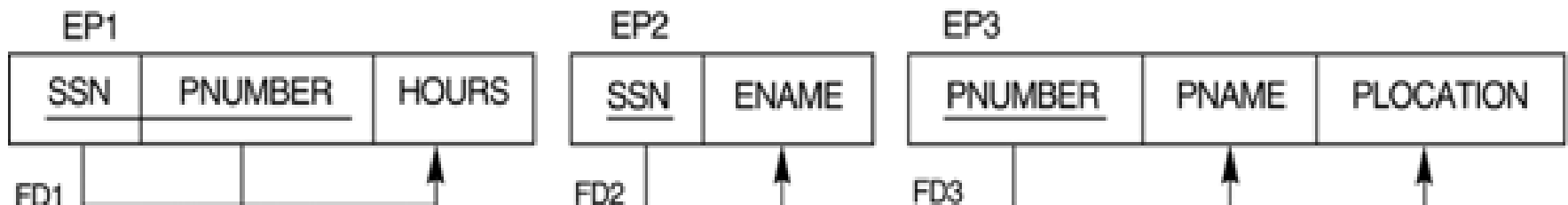
Second Normal Form

- A relation schema R is in **second normal form (2NF)** if every non-prime attribute A in R is fully functionally dependent on the primary key
- R can be decomposed into 2NF relations via the process of 2NF normalization

Second Normal Form



2NF NORMALIZATION



Third Normal Form

Definition:

□ **Transitive functional dependency** - a FD $X \rightarrow Z$

that can be derived from two FDs $X \rightarrow Y$ and $Y \rightarrow Z$

Examples:

- $SSN \rightarrow DMGRSSN$ is a *transitive* FD since

$SSN \rightarrow DNUMBER$ and $DNUMBER \rightarrow DMGRSSN$ hold

- $SSN \rightarrow ENAME$ is *non-transitive* since there is no set

of attributes X where $SSN \rightarrow X$ and $X \rightarrow ENAME$

Third Normal Form

- A relation schema R is in **third normal form (3NF)** if it is in 2NF *and* no non-prime attribute A in R is transitively dependent on the primary key
- R can be decomposed into 3NF relations via the process of 3NF normalization

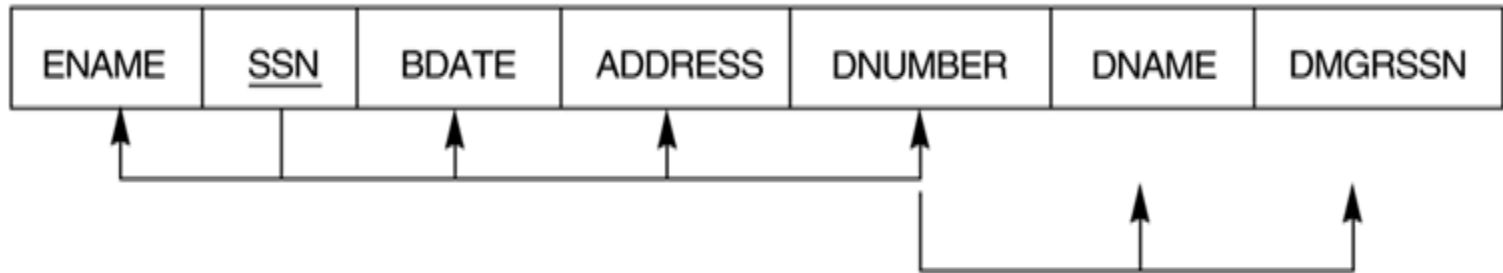
NOTE:

In $X \rightarrow Y$ and $Y \rightarrow Z$, with X as the primary key, we consider this a problem only if Y is not a candidate key. When Y is a candidate key, there is no problem with the transitive dependency .

E.g., Consider EMP (SSN, Emp#, Salary).

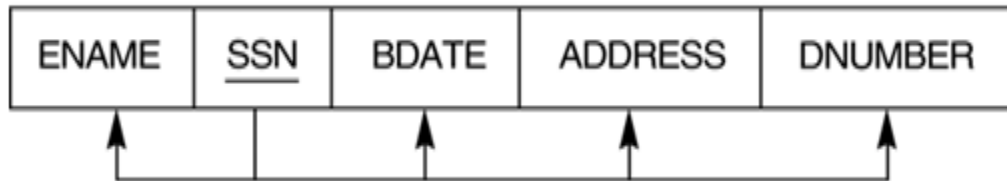
Here, SSN $\rightarrow$ Emp# $\rightarrow$ Salary and Emp# is a candidate key.

EMP_DEPT

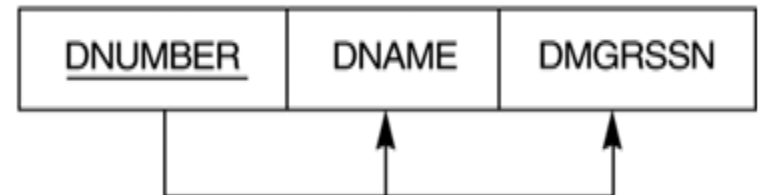


3NF NORMALIZATION

ED1



ED2



BCNF (Boyce-Codd Normal Form)

- A relation schema R is in **Boyce-Codd Normal Form (BCNF)** if whenever an FD $X \rightarrow A$ holds in R, then X is a superkey of R
- Each normal form is strictly stronger than the previous one
 - ▣ Every 2NF relation is in 1NF
 - ▣ Every 3NF relation is in 2NF
 - ▣ Every BCNF relation is in 3NF
- There exist relations that are in 3NF but not in BCNF
- The goal is to have each relation in BCNF (or 3NF)

LOTS1A

<u>PROPERTY_ID#</u>	COUNTY_NAME	LOT#	AREA
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FD1

FD2

FD5

BCNF Normalization

LOTS1AX

<u>PROPERTY_ID#</u>	AREA	LOT#
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LOTS1AY

<u>AREA</u>	COUNTY_NAME
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FIGURE 10.13

A relation TEACH that is in 3NF but not BCNF.

TEACH

STUDENT	COURSE	INSTRUCTOR
Narayan	Database	Mark
Smith	Database	Navathe
Smith	Operating Systems	Ammar
Smith	Theory	Schulman
Wallace	Database	Mark
Wallace	Operating Systems	Ahamad
Wong	Database	Omiecinski
Zelaya	Database	Navathe